

RAG → Retrieval Augmented Generation.

→ It is process of optimizing the o/p of LLM, so it references an authoritative knowledge based outside of its training data source before generating a response.

→ LLM are trained on vast volumes of data and use billions of parameters to generate original o/p for tasks like Q/A, translating languages and completing sentences.

→ RAG extends the already powerful capabilities of LLMs to specific domains or an organization's internal knowledge base, all without the need to retrain the model.

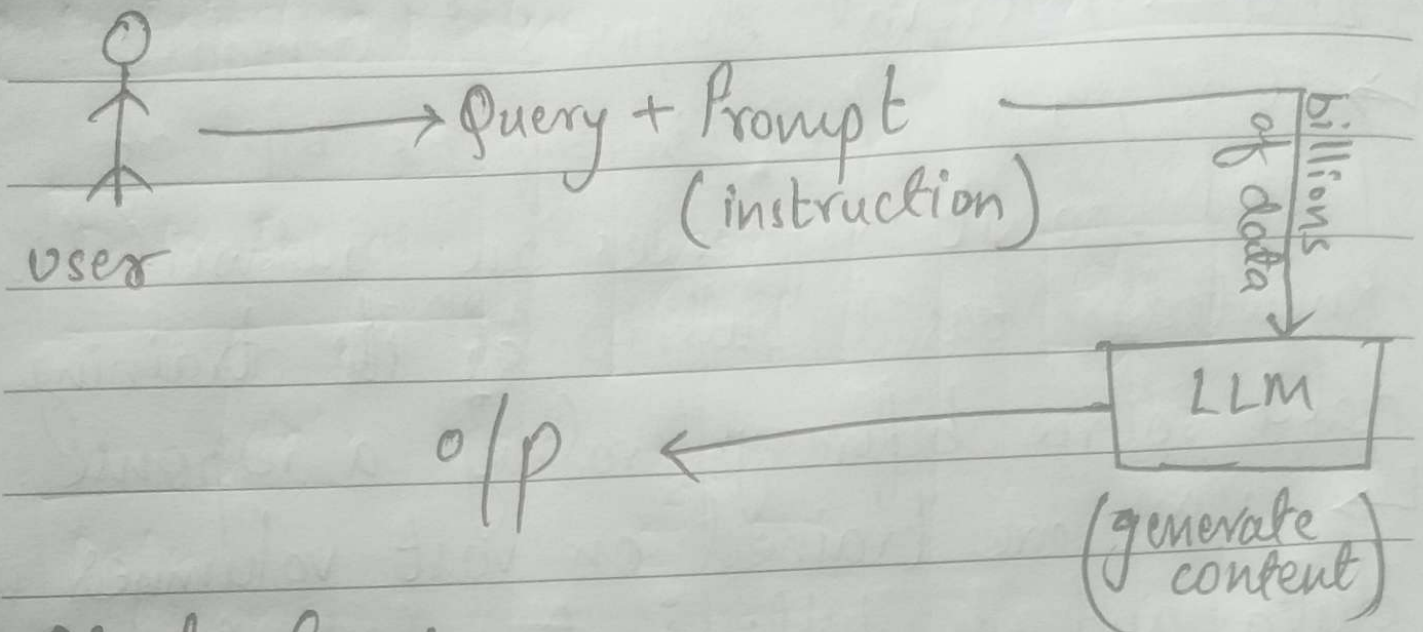
→ It is cost-effective approach to improve or improving LLM o/p so it remains relevant, accurate, and useful in various contexts.



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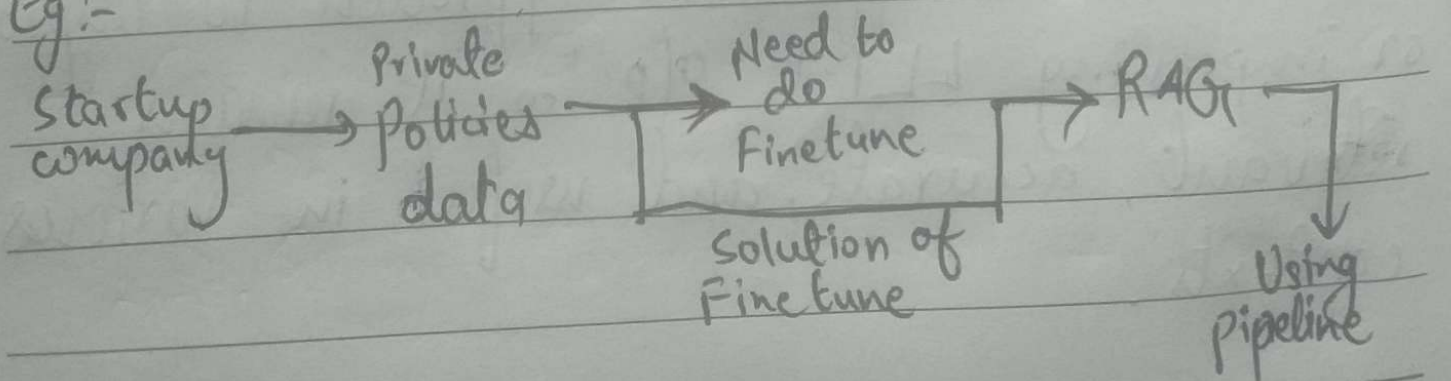
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Disadvantages

- Trained for specific purpose only. (LLM).
- Hallucinating.
 - i.e, only have specific time period of data no latest data and its o/p.
- Private data need to do Finetune which is huge time consuming in LLM. So, we use RAG with the help of pipeline.

Eg:-



Note:-

- External DB has different data in diff. format.
- Data are use for parsing then embedding.
- Embedding means "text + Vectors".

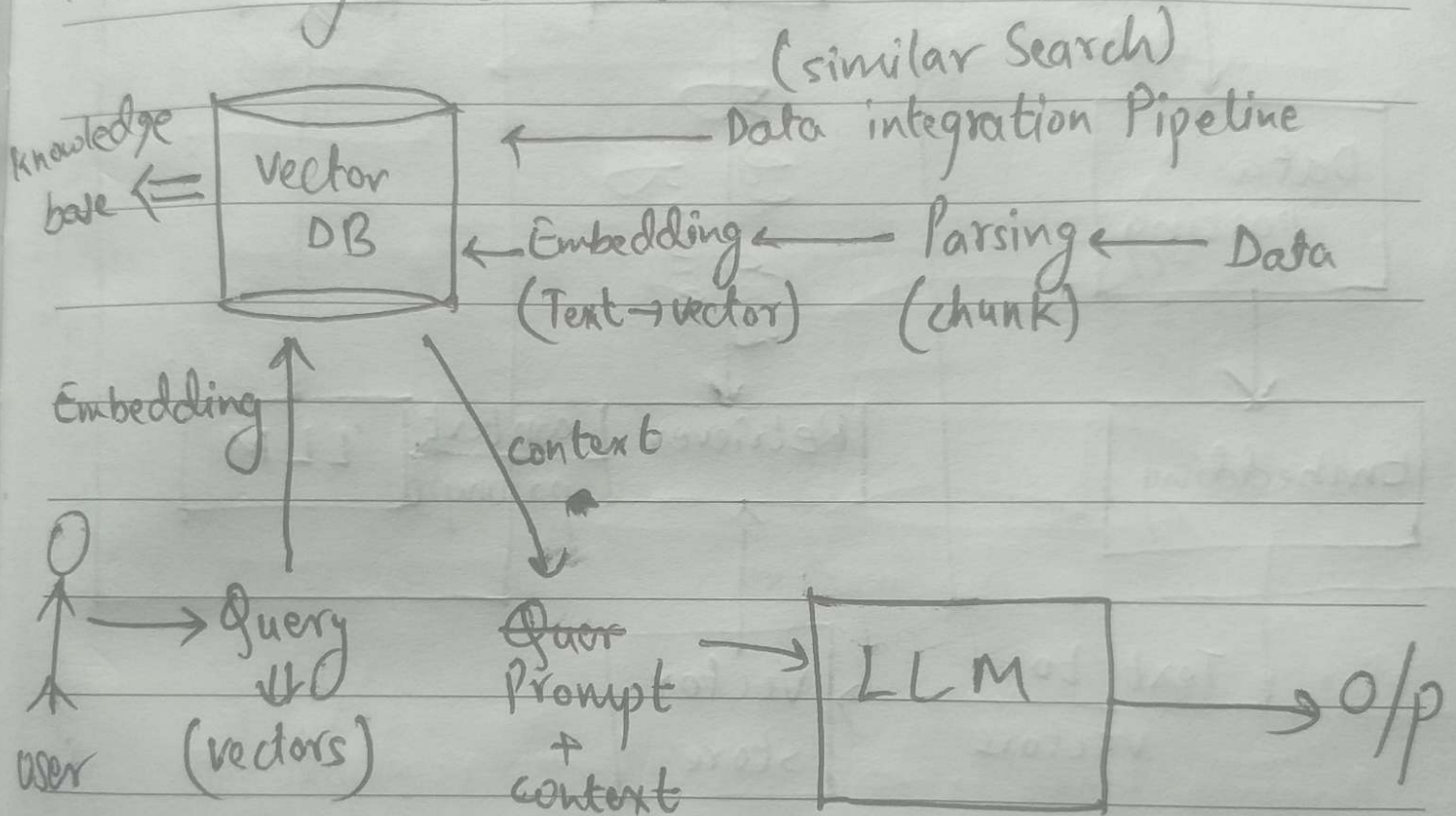


Fig:- Retrieval Pipeline [Traditional RAG]

Notes:-

- Perplexity is build by using RAG + Tools.



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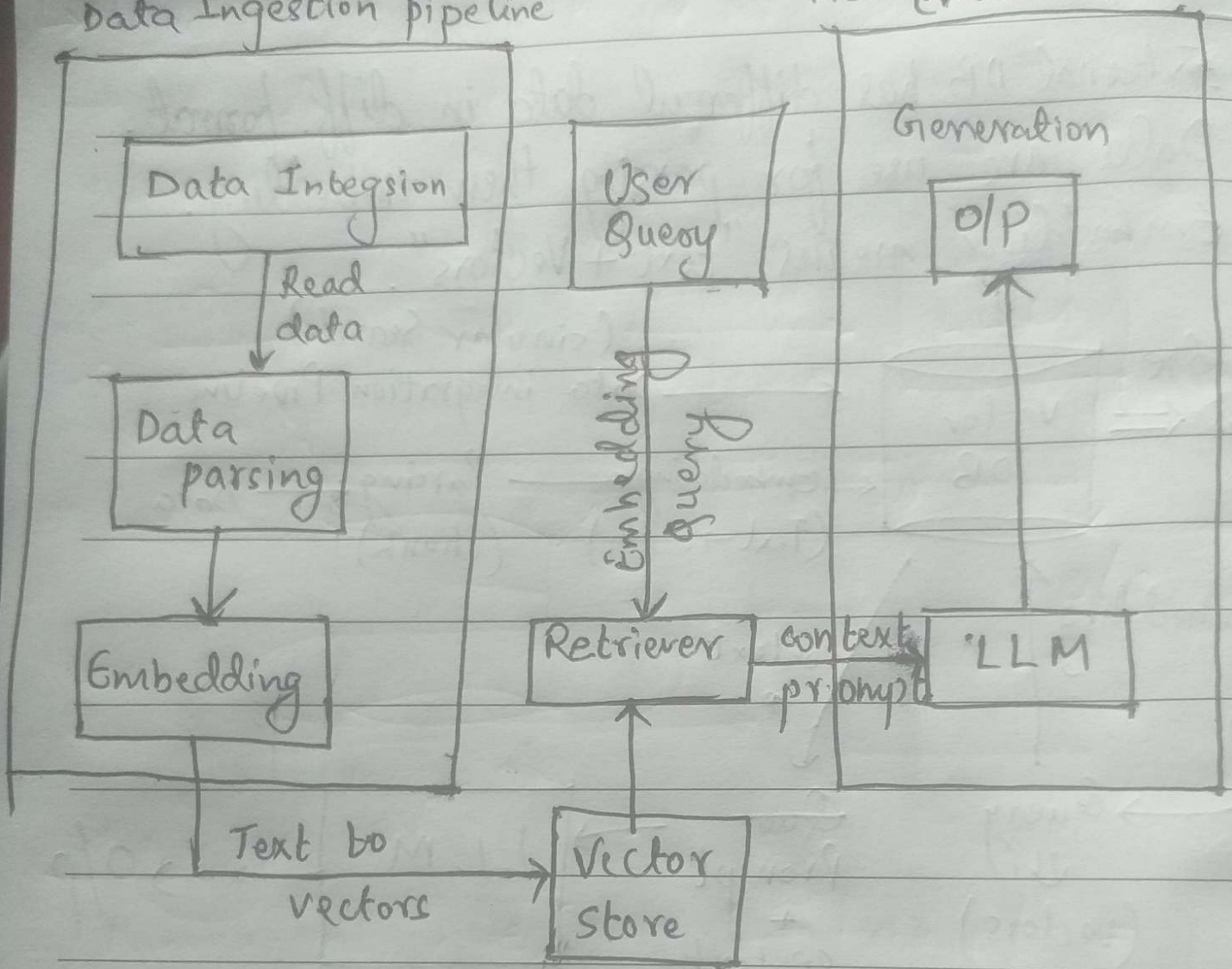
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Data Ingestion pipeline

Retrieval Pipeline



Agenda

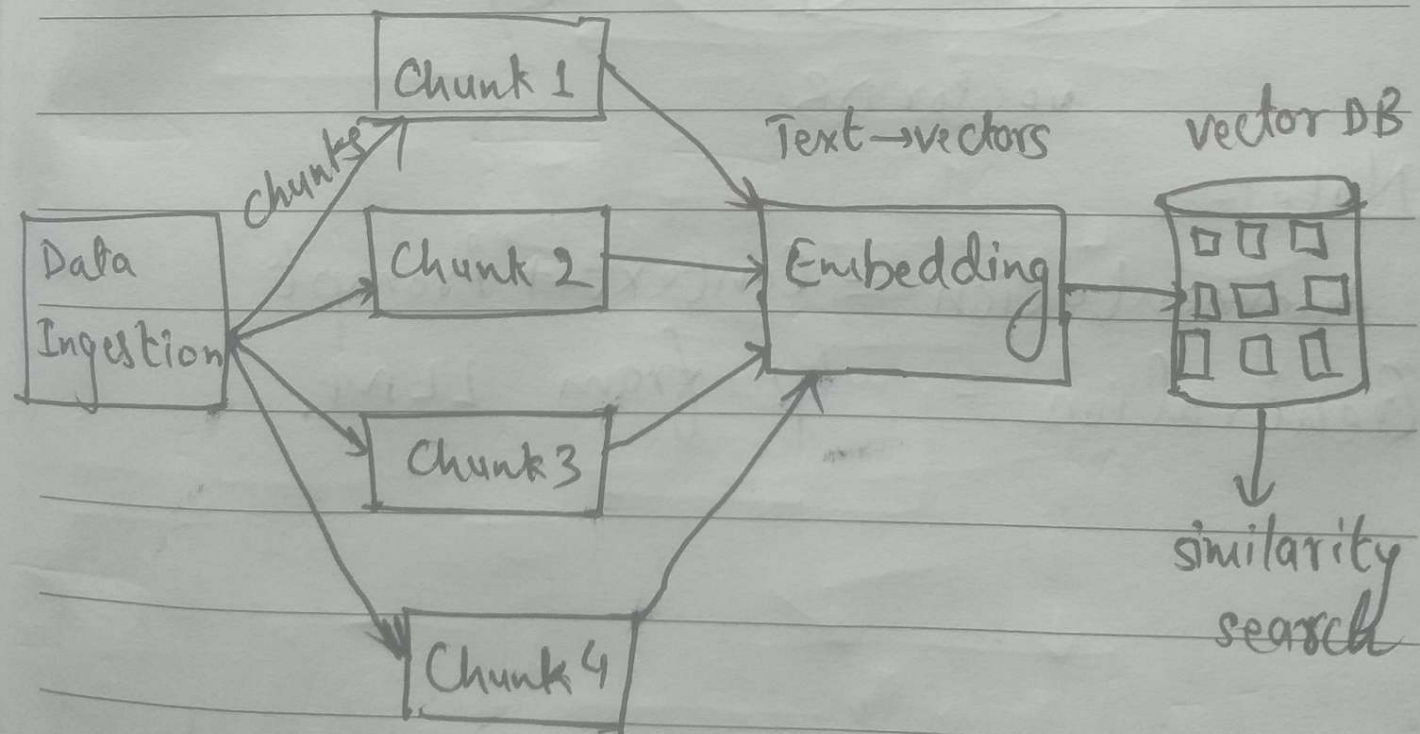
→ 1). Document Structure

2). Complete RAG Pipeline

2.1). Data Ingestion Pipeline

2.2). Query Retrieval Pipeline + LLM

- Here, inside document structure there is presence of Metadata, content, structure, etc.
- Data parsing is used to chunk the huge data in small and equal size of content vector.
having ~~the~~ same context size.
- We store data in vector DB for similarity search.
- In data Ingestion has huge data in diff. format so we used Document structure with chunk methods.

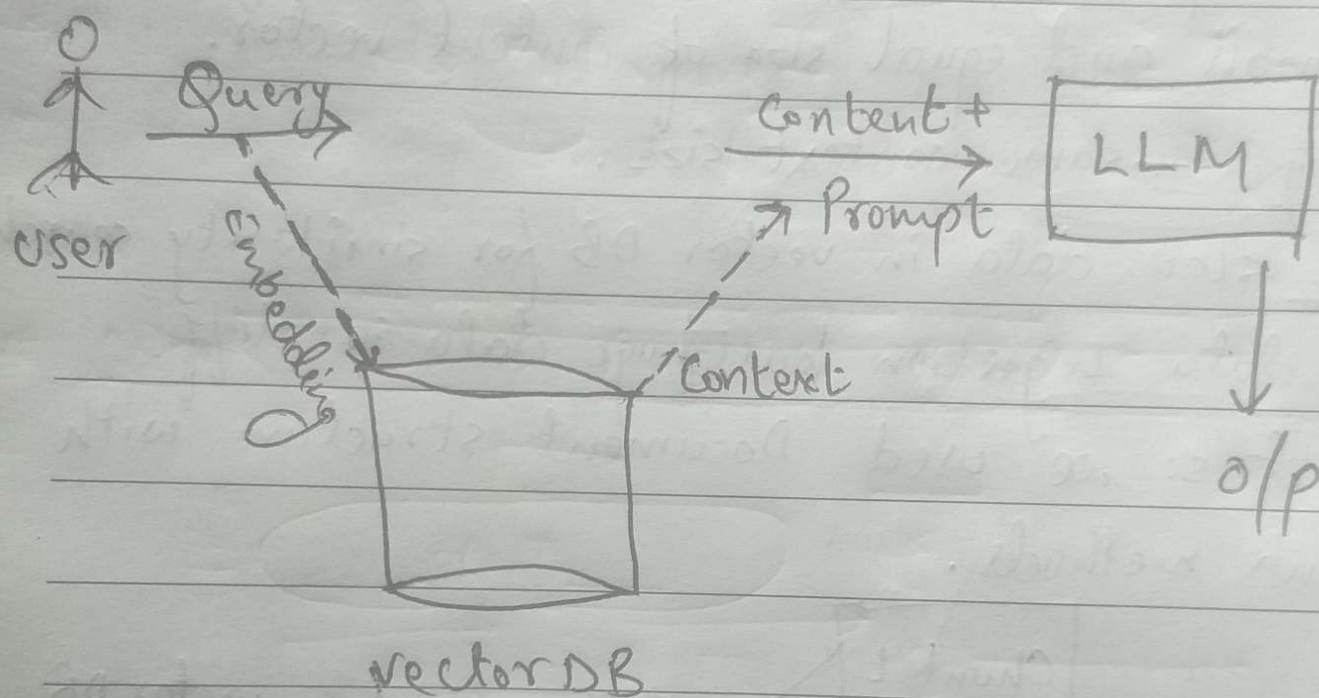




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→ RAG is commonly known as Augmented Generation.



Note:-

Augmentation = Context + Prompt
Generation = o/p from LLM.



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AGENTIC RAG

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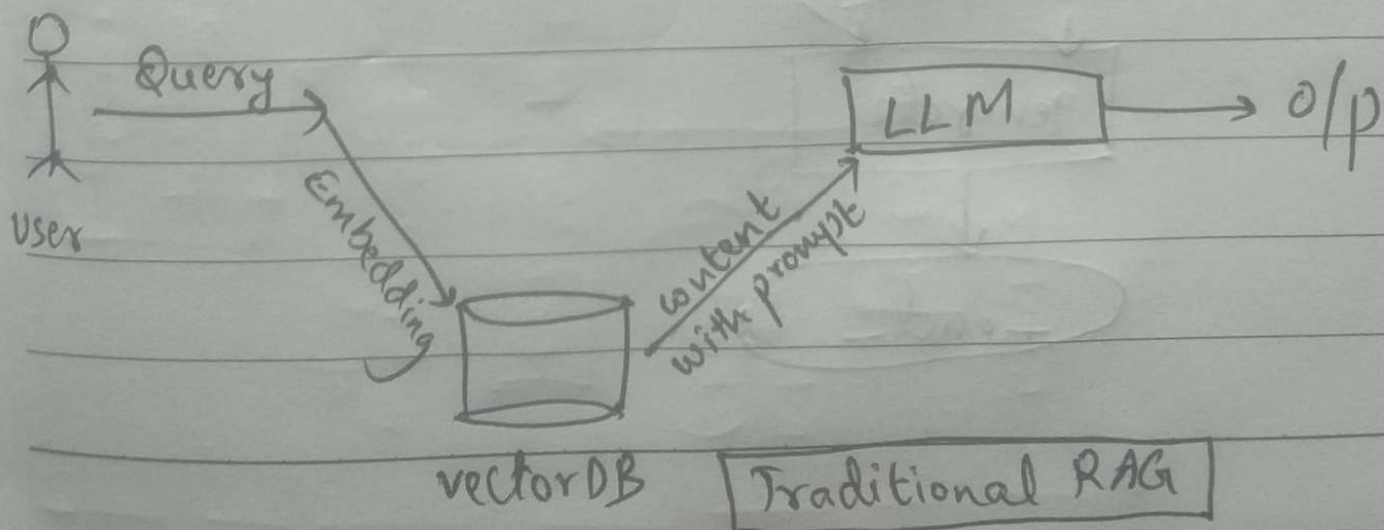
- ↳ Combination of 2 important components
- RAG (Retrieval Augmented Generation)
 - Autonomous Agent

RAG + Autonomous Agent

- Intelligent, dynamic Decisions about the retrieval and Generation.
- ~~External RAG~~ ⇒ External Data Source

↳ RAG = External API, Tool, or Vector DB, Databases.
or Data Sources.

↳ Autonomous Agent = Intelligent, dynamic Decisions about the retrieval and generation.





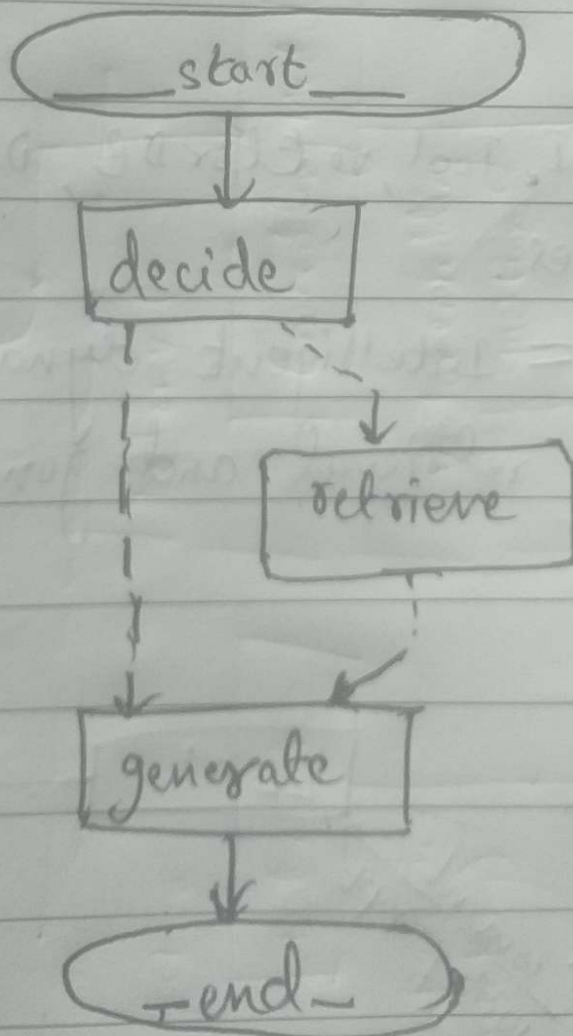
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- # What Autonomous Agent do?
- a). when to retrieve information,
b). what to retrieve
c). where to retrieve
d). How many times to retrieve

WorkFlow of AGENTIC RAG

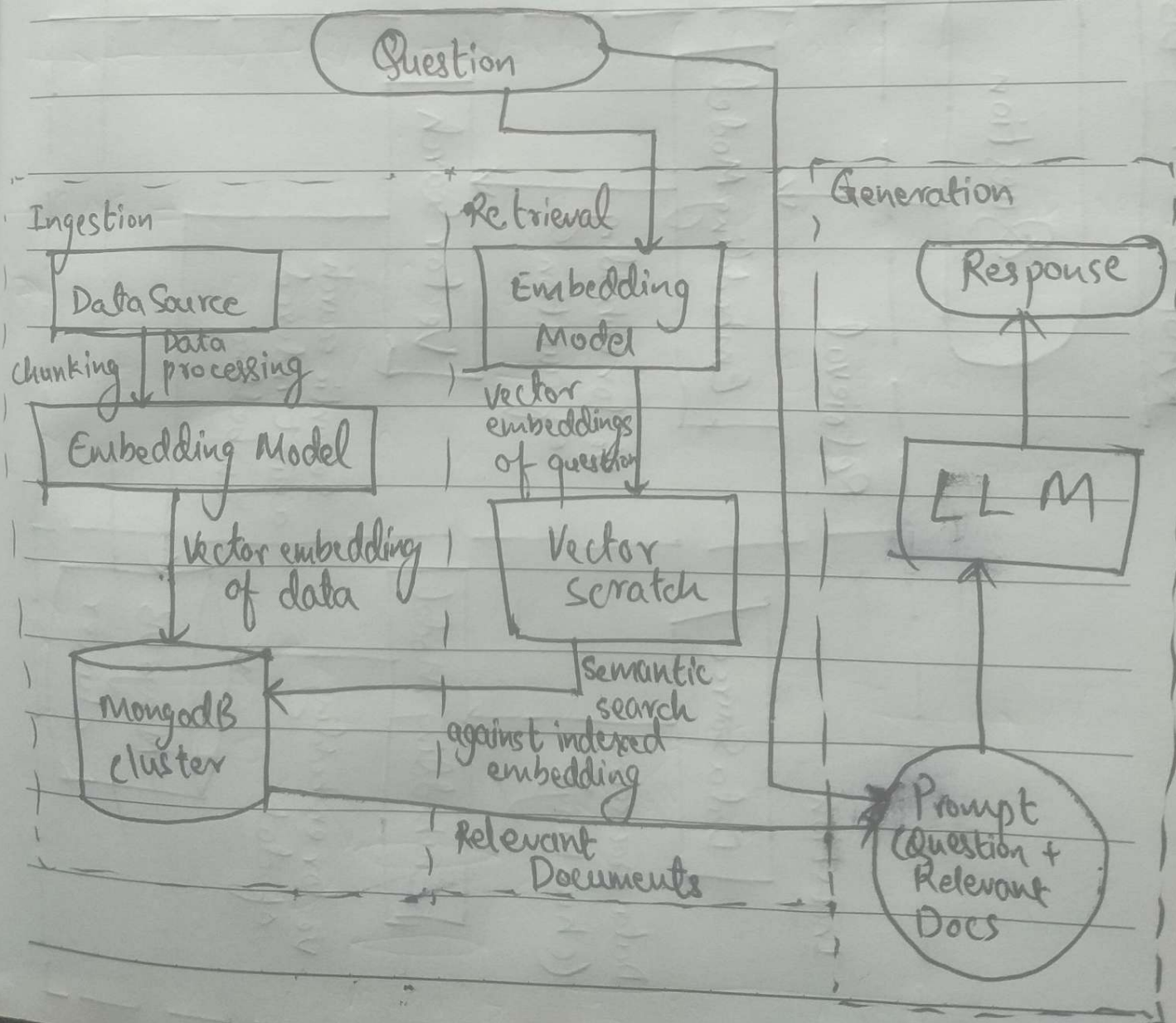




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- We implement workflow by using langchain.
- Decide act as agent
- retrieve act as vector DB
- Generate act as LLM.

RAG with MongoDB Vector Search Search:-



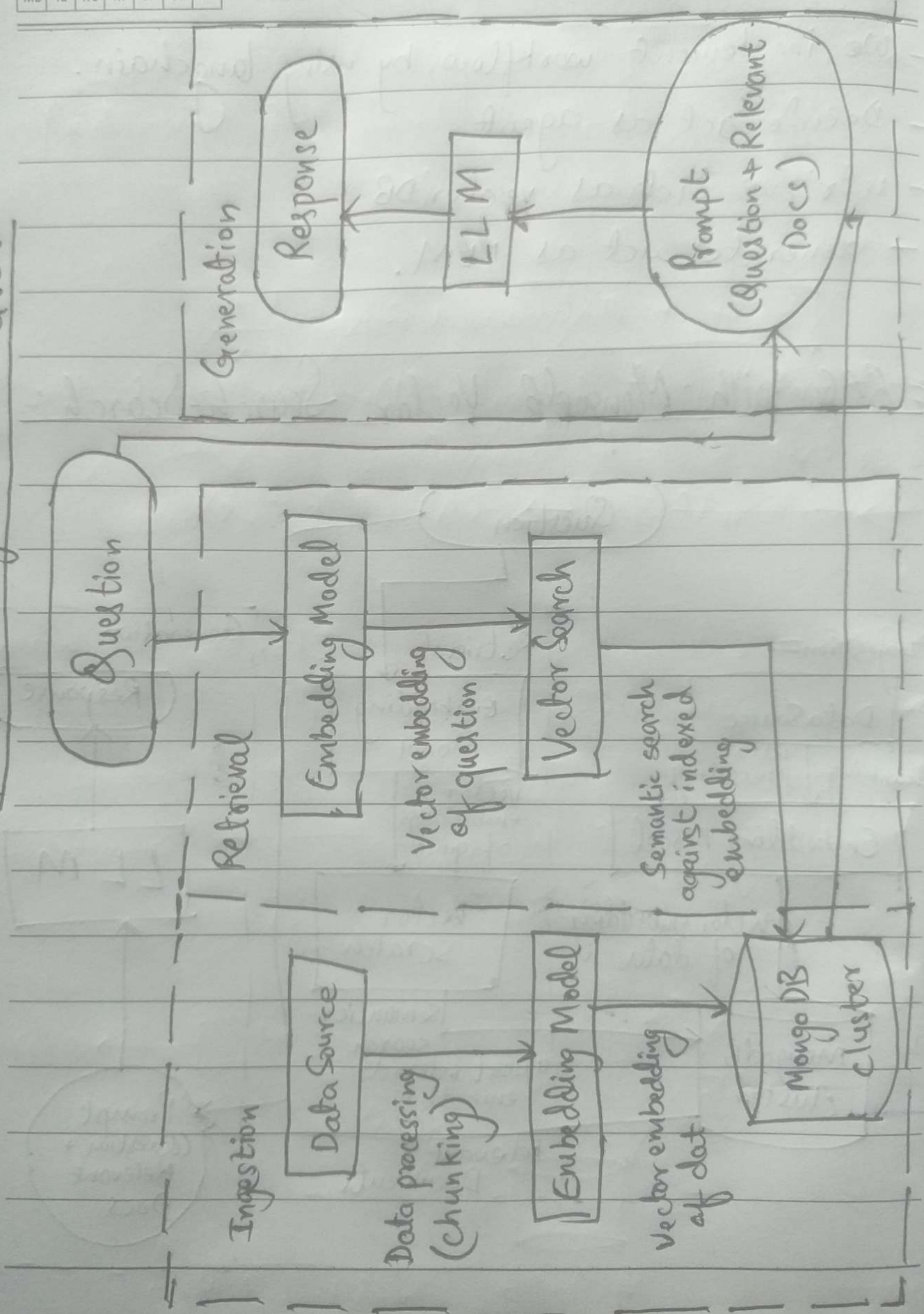
RAG with MongoDB Vector Search



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Evaluation of RAG Application & Chatbot.



→ We need to have knowledge of

a). which LLM model is used

b). Ground Truth of o/p

c). How to create the data.

d). Evaluation of matrix from data.

eg:- LLM as judge + Prompt.

~~e). Gathering of data points.~~

~~f). Un.~~

For CHATBOT → By using LangSmith

a). Gathering data points

b). LLM as JUDGE

c). Evaluation matrix

d). Comparison with multiple LLM models to select.

for tracking
in cloud